Deploy a Jenkins CI/CD Pipeline with Persistent Storage: Red Hat OpenShift with NetApp

NetApp Solutions

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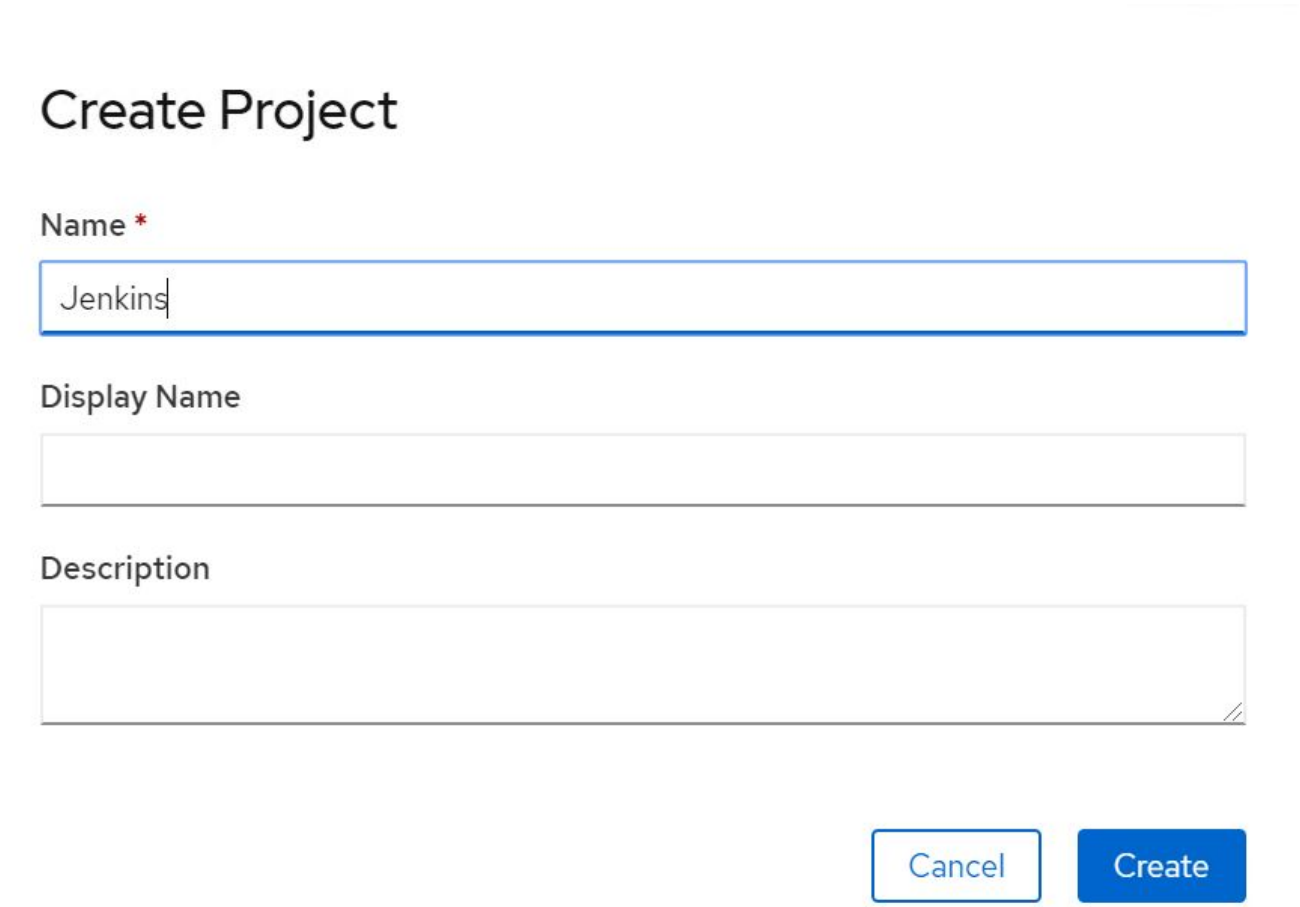
Deploy a Jenkins CI/CD Pipeline with Persistent Storage: Red Hat OpenShift with NetApp

This section provides the steps to deploy a continuous integration/continuous delivery or deployment (CI/CD) pipeline with Jenkins to validate solution operation.

Create the resources required for Jenkins deployment

To create the resources required for deploying the Jenkins application, complete the following steps:

1. Create a new project named Jenkins.



2. In this example, we deployed Jenkins with persistent storage. To support the Jenkins build, create the PVC. Navigate to `Storage > Persistent Volume Claims` and click `Create Persistent Volume Claim`. Select the storage class that was created, make sure that the Persistent Volume Claim Name is jenkins, select the appropriate size and access mode, and then click Create.

Project: jenkins

Create Persistent Volume Claim

Edit YAML

Storage Class

SC basic

Storage class for the new claim.

Persistent Volume Claim Name *

jenkins

A unique name for the storage claim within the project.

Access Mode *

Single User (RWO) Shared Access (RWX) Read Only (ROX)

Permissions to the mounted drive.

Size *

100 GiB

Desired storage capacity.

Use label selectors to request storage

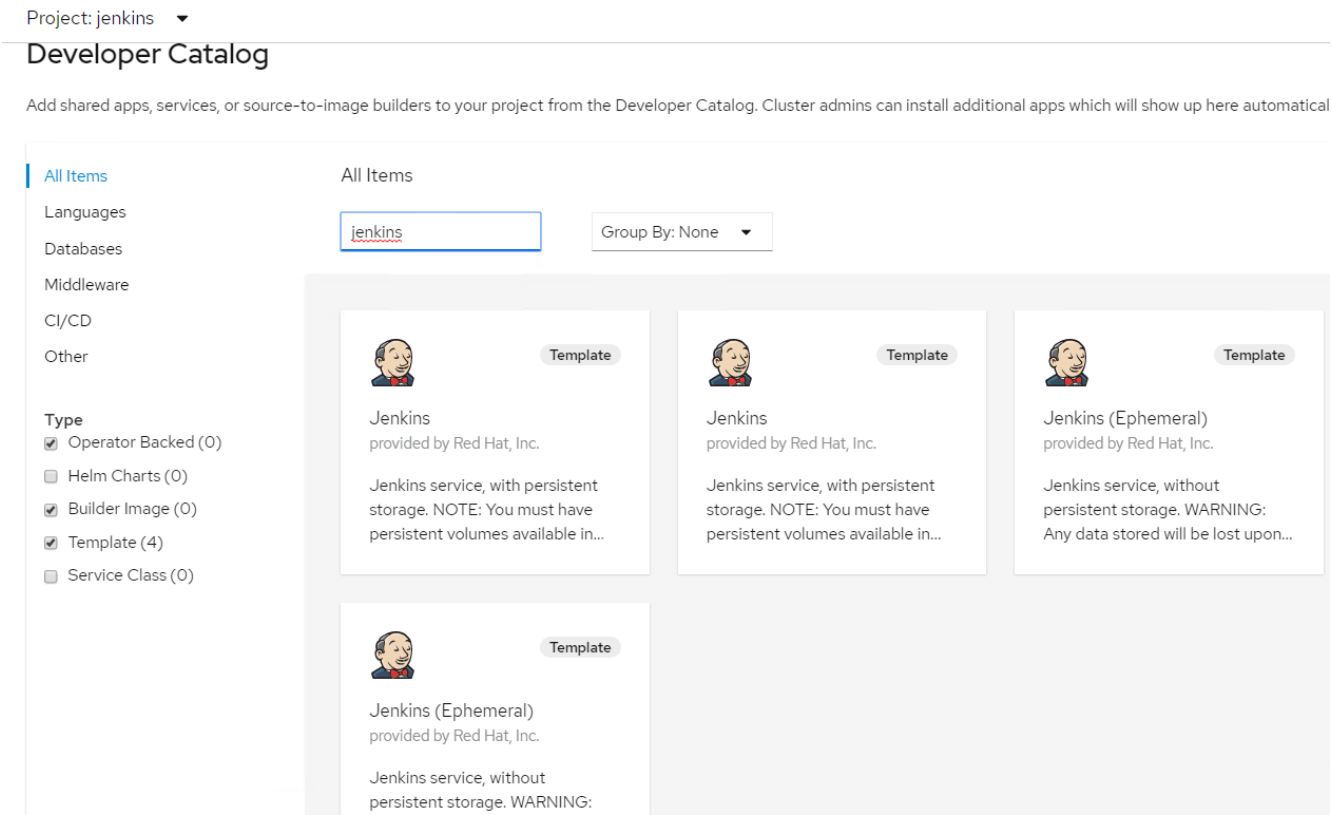
Use label selectors to define how storage is created.

Create Cancel

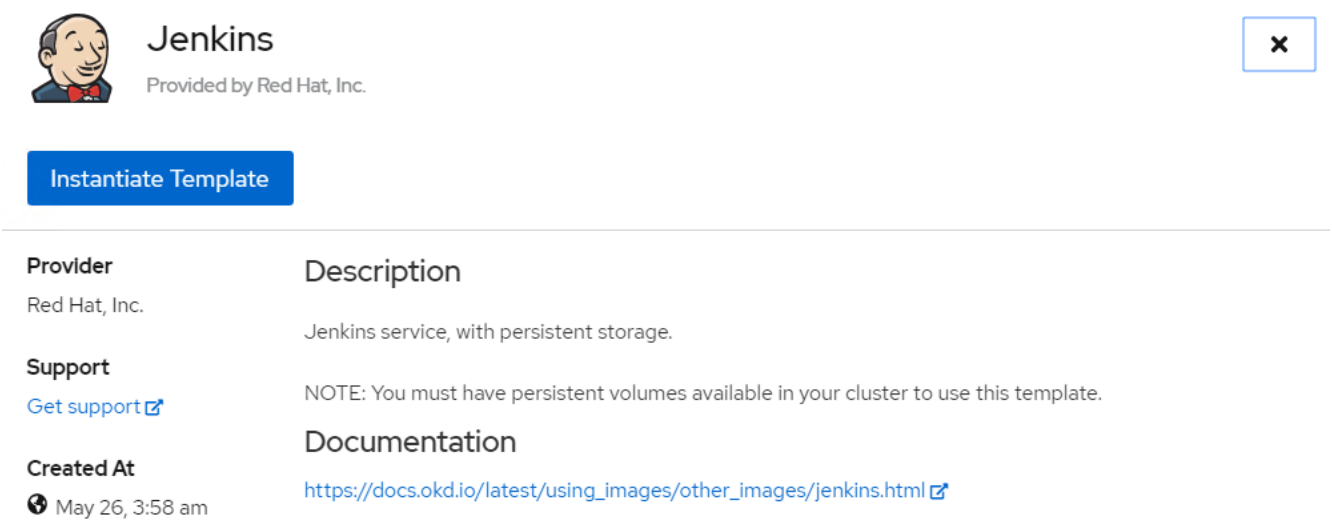
Deploy Jenkins with Persistent Storage

To deploy Jenkins with persistent storage, complete the following steps:

1. In the upper left corner, change the role from Administrator to Developer. Click `+Add` and select `From Catalog`. In the `Filter by Keyword` bar, search for jenkins. Select Jenkins Service with Persistent Storage.



2. Click `Instantiate Template`.



3. By default, the details for the Jenkins application are populated. Based on your requirements, modify the parameters and click Create. This process creates all the required resources for supporting Jenkins on

OpenShift.

Instantiate Template

Namespace *

PR jenkins

Jenkins Service Name

jenkins

The name of the OpenShift Service exposed for the Jenkins container.

Jenkins JNLP Service Name

jenkins-jnlp

The name of the service used for master/slave communication.

Enable OAuth in Jenkins

true

Whether to enable OAuth OpenShift integration. If false, the static account 'admin' will be initialized with the password 'password'.

Memory Limit

Gi

Maximum amount of memory the container can use.

Volume Capacity *

50Gi

Volume space available for data, e.g. 512Mi, 2Gi.

Jenkins ImageStream Namespace

openshift

The OpenShift Namespace where the Jenkins ImageStream resides.

Disable memory intensive administrative monitors

false

Whether to perform memory intensive, possibly slow, synchronization with the Jenkins Update Center on start. If true, the Jenkins core update monitor and site warnings monitor are disabled.

Jenkins ImageStreamTag

jenkins:2

Name of the ImageStreamTag to be used for the Jenkins image.

Fatal Error Log File

false

When a fatal error occurs, an error log is created with information and the state obtained at the time of the fatal error.

Allows use of Jenkins Update Center repository with invalid SSL certificate

false

Whether to allow use of a Jenkins Update Center that uses invalid certificate (self-signed, unknown CA). If any value other than 'false', certificate check is bypassed. By default, certificate check is enforced.

Create Cancel

Jenkins

INSTANT-APP JENKINS

View documentation Get support

Jenkins service, with persistent storage.

NOTE: You must have persistent volumes available in your cluster to use this template.

The following resources will be created:

- DeploymentConfig
- PersistentVolumeClaim
- RoleBinding
- Route
- Service
- ServiceAccount

4. The Jenkins pods take approximately 10–12 minutes to enter the Ready state.

Project: jenkins

Pods

Create Pod

Filter by name...

1 Running | 0 Pending | 0 Terminating | 0 CrashLoopBackOff | 1 Completed | 0 Failed | 0 Unknown

Select all filters

1 of 2 Items

Name	Namespace	Status	Ready	Owner	Memory	CPU
P jenkins-1-c77n9	NS jenkins	Running	1/1	RC jenkins-1	-	0.004 cores

5. After the pods are instantiated, navigate to `Networking > Routes`. To open the Jenkins webpage, click the URL provided for the jenkins route.

Project: jenkins

Routes

Create Route

Filter by name...

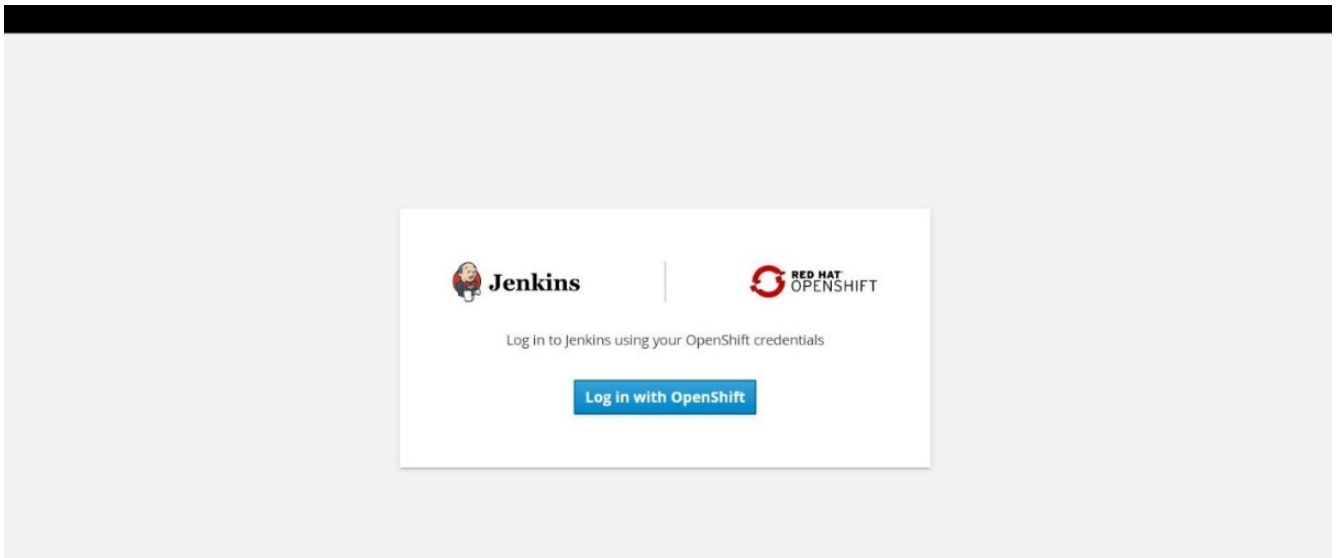
1 Accepted | 0 Rejected | 0 Pending

Select all filters

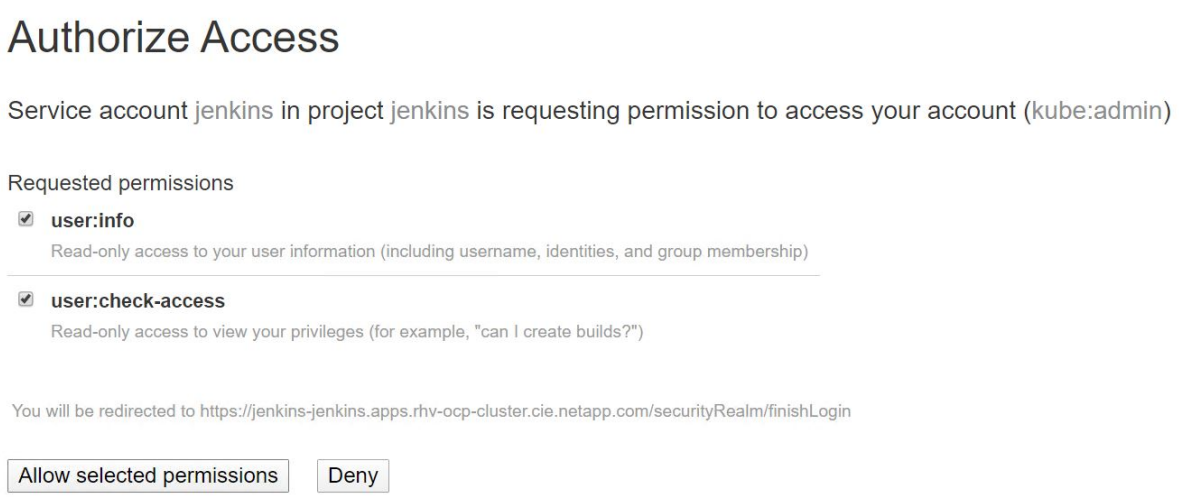
1 Item

Name	Namespace	Status	Location	Service
RT jenkins	NS jenkins	Accepted	https://jenkins-jenkins.apps.rhv-ocp-cluster.cie.netapp.com	S jenkins

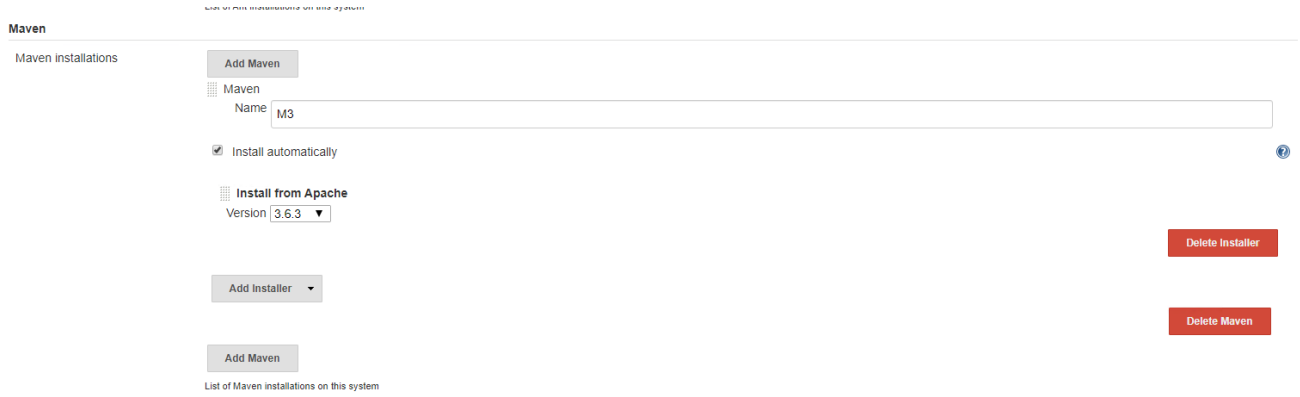
6. Because OpenShift OAuth was used while creating the Jenkins app, click `Log in with OpenShift`.



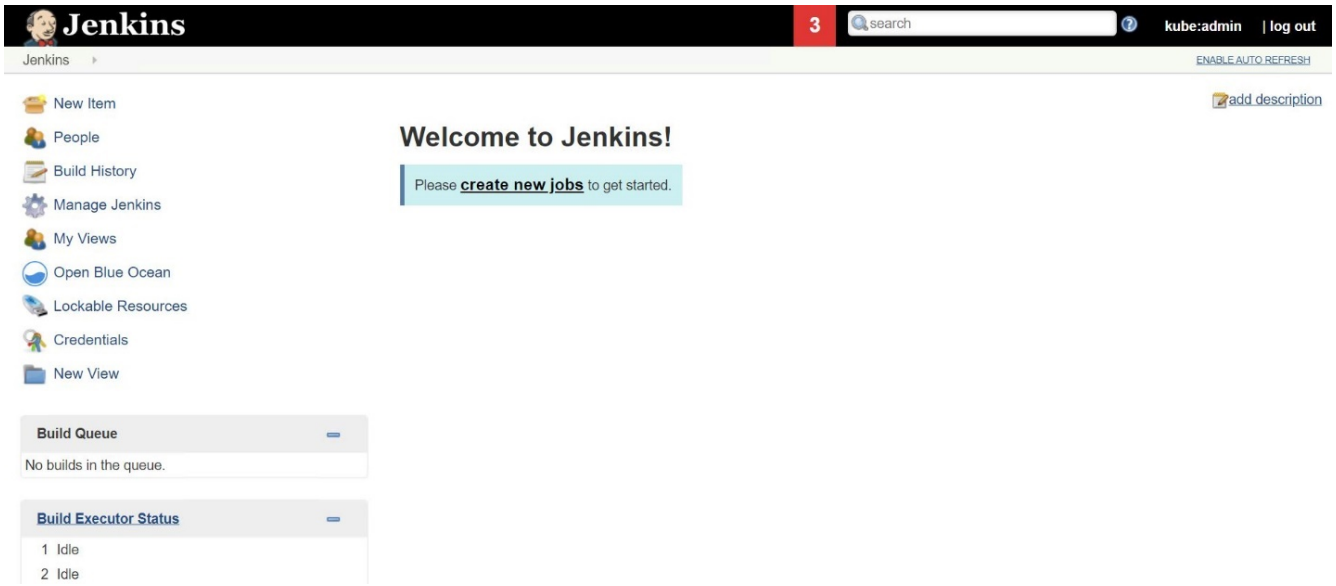
7. Authorize the Jenkins service-account to access the OpenShift users.



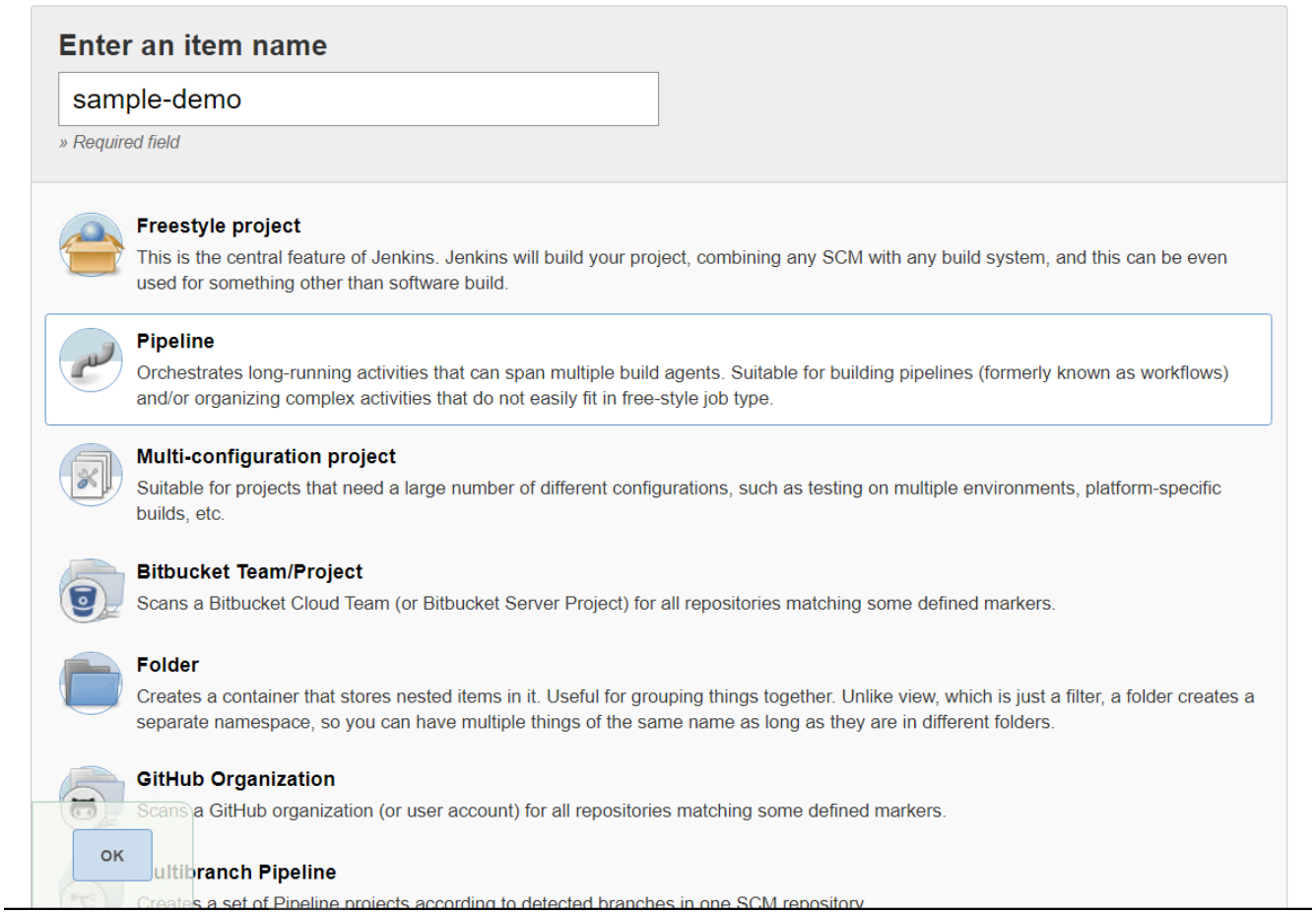
8. The Jenkins welcome page is displayed. Because we are using a Maven build, complete the Maven installation first. Navigate to `Manage Jenkins > Global Tool Configuration`, and then, in the Maven subhead, click `Add Maven`. Enter the name of your choice and make sure that the Install Automatically option is selected. Click Save.



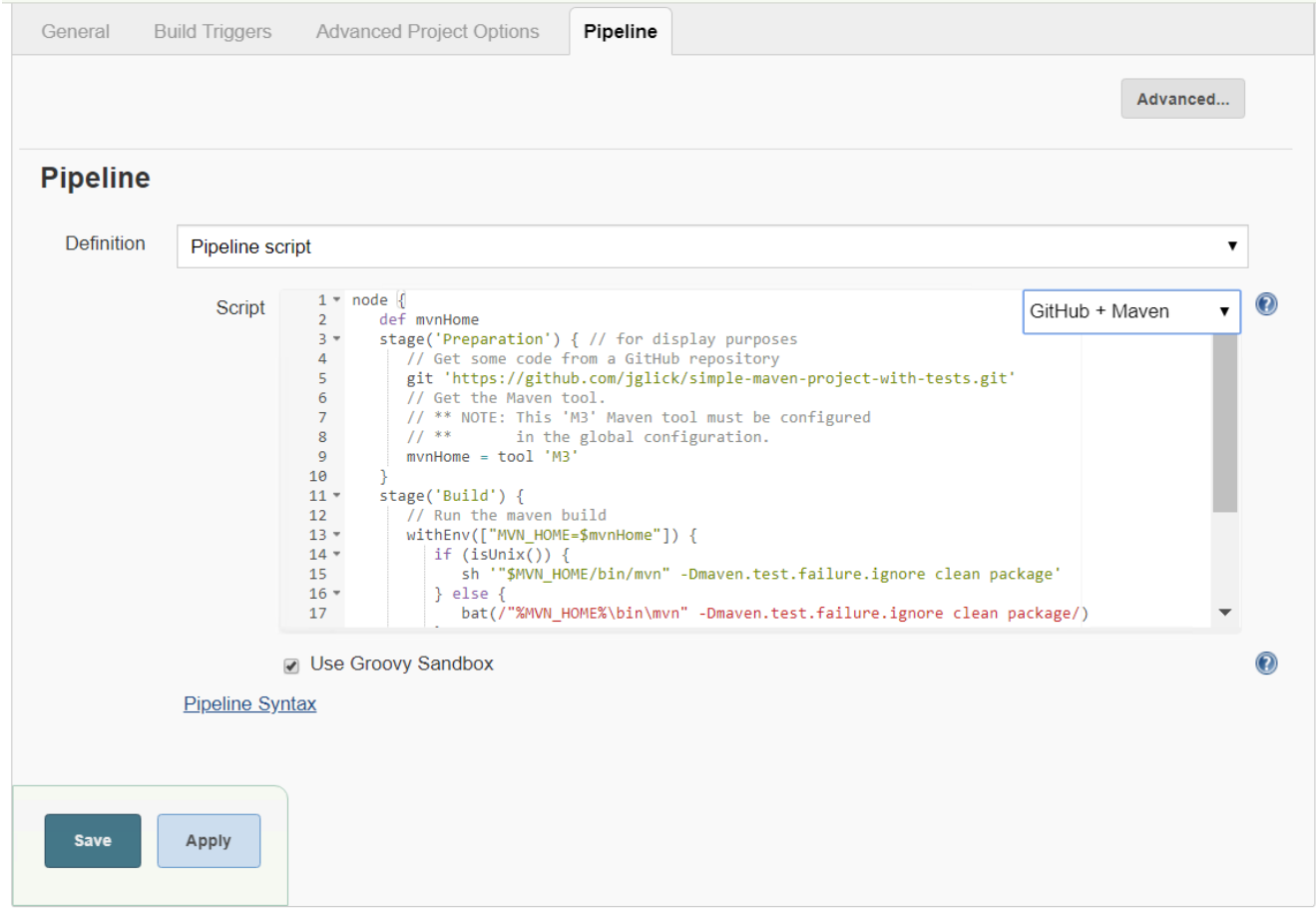
9. You can now create a pipeline to demonstrate the CI/CD workflow. On the home page, click `Create New Jobs` or `New Item` from the left-hand menu.



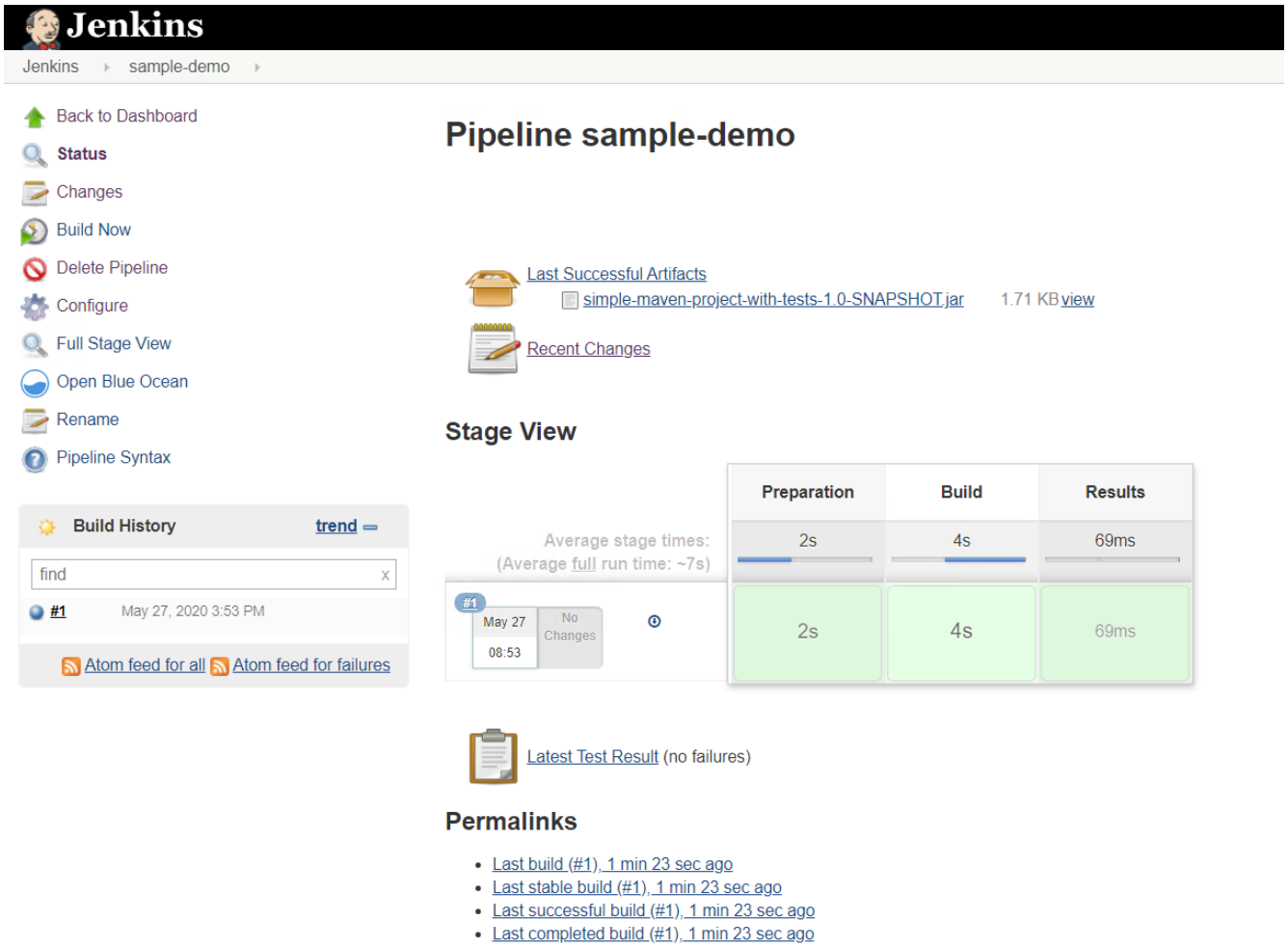
10. On the Create Item page, enter the name of your choice, select Pipeline, and click Ok.



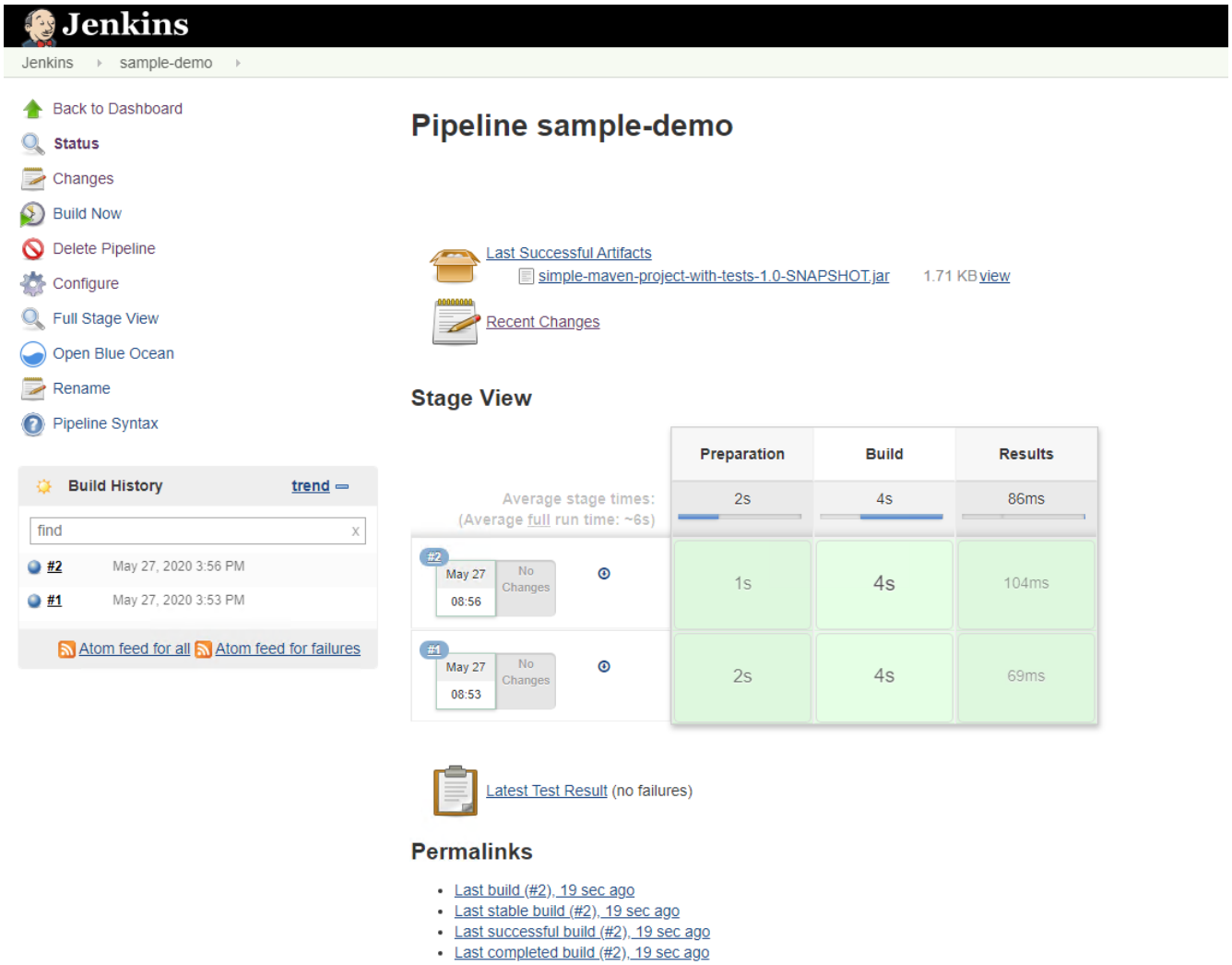
11. Select the Pipeline tab. From the Try Sample Pipeline drop-down menu, select `Github + Maven`. The code is automatically populated. Click Save.



12. Click `Build Now` to trigger the development through the preparation, build, and testing phase. It can take several minutes to complete the whole build process and display the results of the build.



13. Whenever there are any code changes, the pipeline can be rebuilt to patch the new version of software enabling continuous integration and continuous delivery. Click `Recent Changes` to track the changes from the previous version.



Next: Videos and Demos.

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